

California Housing Price Prediction

A Linear Regression Project

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Batch: A

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Project Type: Linear Regression (Individual Project)

Dataset: California Housing Dataset

Dataset Source: GeeksforGeeks - Dataset for Linear Regression

Code Notebook (Colab): <https://colab.research.google.com/drive/14IMW6ggL0d-o7hSHwVLkgnXvIL4TJI6F?usp=sharing>

2. Objective

Build a linear regression model to predict the median house value (`median_house_value`) of a California district using housing and demographic features - location (longitude, latitude), housing age, room/bedroom counts, population, households, median income, and proximity to the ocean. The goal is to understand which factors most strongly influence housing prices and to evaluate how accurately a linear model can estimate house value on unseen data (measured via MSE, MAE, and R-squared).

3. Dataset Overview

The dataset is the California Housing dataset, collected from the 1990 U.S. Census. It contains 20,640 rows and 10 columns describing housing and demographic attributes for California districts.

Property	Value
Rows	20,640
Columns	10
Target variable	<code>median_house_value</code> (float64)
Numeric features (float64)	<code>longitude</code> , <code>latitude</code> , <code>housing_median_age</code> , <code>total_rooms</code> , <code>total_bedrooms</code> , <code>population</code> , <code>households</code> , <code>median_income</code>
Categorical feature (object)	<code>ocean_proximity</code> (5 categories: <code><1H OCEAN</code> , <code>INLAND</code> , <code>NEAR OCEAN</code> , <code>NEAR BAY</code> , <code>ISLAND</code>)

Sample records (first 3 rows):

<code>longitude</code>	<code>latitude</code>	<code>housing_median_age</code>	<code>total_rooms</code>	<code>median_income</code>	<code>median_house_value</code>	<code>ocean_proximity</code>
-122.23	37.88	41.0	880.0	8.33	452600.0	NEAR BAY
-122.22	37.86	21.0	7099.0	8.30	358500.0	NEAR BAY
-122.24	37.85	52.0	1467.0	7.26	352100.0	NEAR BAY

4. Data Quality and Cleaning

Check	Before Cleaning	After Cleaning
Rows	20,640	20,640
Duplicate rows	0	0
Missing values (total_bedrooms)	207	0
ocean_proximity data type	text (5 categories)	encoded as integer (0-4)

Outlier check:

The boxplot of numeric features (see Section 6) shows several points above the upper whisker for total_rooms, total_bedrooms, and population. These were treated as genuine high-density districts rather than data errors and were not removed, since removing them would discard real variation the model needs to learn from.

Cleaning decisions:

- No duplicate rows were found, so no rows were dropped for duplication.
- The total_bedrooms column had 207 missing values (about 1% of the data), filled with the column median so all 20,640 rows are retained (type correction: no dtype change needed, stayed float64).
- ocean_proximity (categorical/text) was label-encoded into integers (0-4) so it could be used as a numeric model input.

5. Descriptive Analysis

Summary statistics for numerical variables:

Feature	Mean	Std Dev	Min	25%	50%	75%	Max
housing_median_age	28.64	12.59	1.0	18.0	29.0	37.0	52.0
total_rooms	2,635.8	2,181.6	2.0	1,447.8	2,127.0	3,148.0	39,320.0
total_bedrooms	537.9	421.4	1.0	296.0	435.0	647.0	6,445.0
population	1,425.5	1,132.5	3.0	787.0	1,166.0	1,725.0	35,682.0
households	499.5	382.3	1.0	280.0	409.0	605.0	6,082.0
median_income	3.87	1.90	0.50	2.56	3.53	4.74	15.00
median_house_value	206,855.8	115,395.6	14,999.0	119,600.0	179,700.0	264,725.0	500,001.0

Median house values range from \$14,999 to \$500,001, with an average of about \$206,856. Median income averages 3.87 (tens of thousands of dollars) and is right-skewed - most districts fall between roughly 2.56 and 4.74. Room, bedroom, population, and household counts are all right-skewed with long tails of high-density districts.

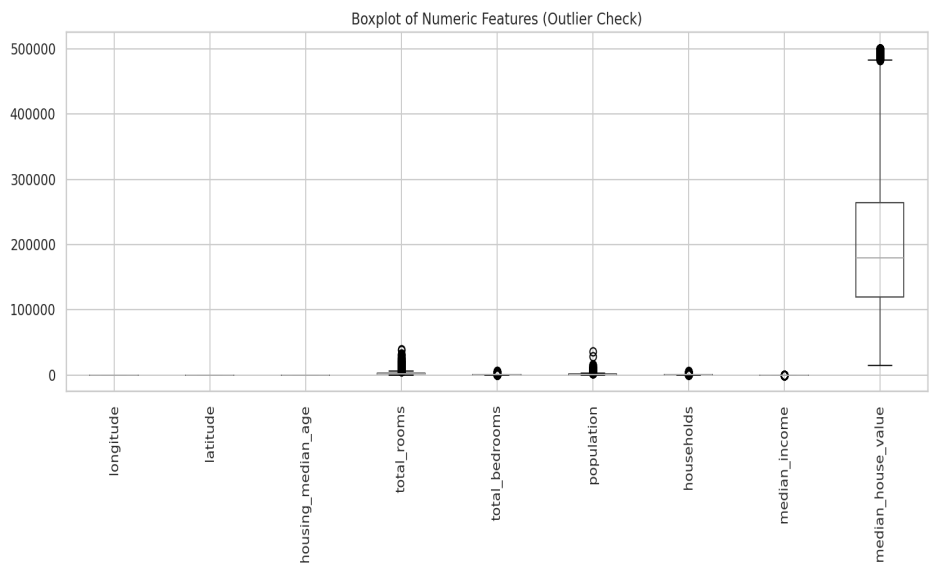
Summary for categorical variable (ocean_proximity):

Category	Count	% of Data
<1H OCEAN	9,136	44.3%
INLAND	6,551	31.7%
NEAR OCEAN	2,658	12.9%
NEAR BAY	2,290	11.1%
ISLAND	5	0.02%

Most districts fall in the '<1H OCEAN' and 'INLAND' categories, together covering about 76% of the data. 'ISLAND' is a very rare category with only 5 records.

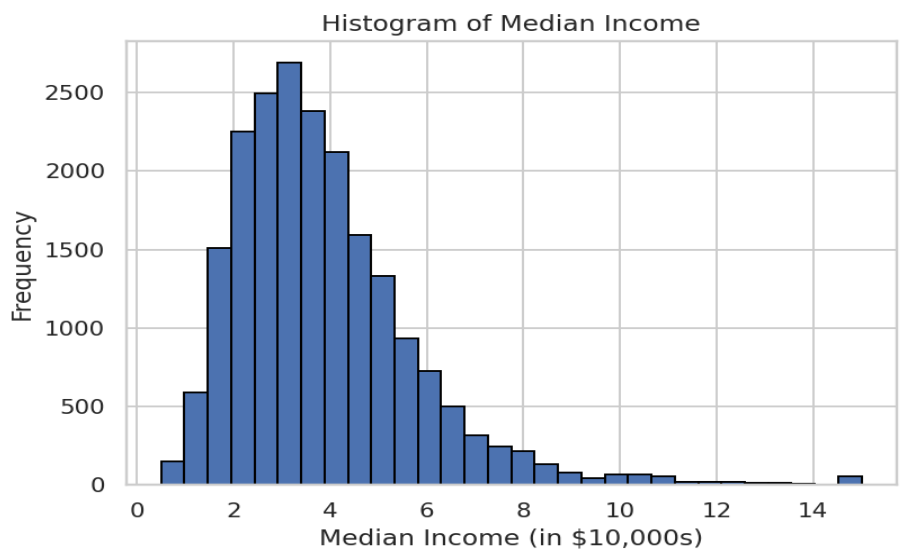
6. Visual Analysis

Chart 1: Boxplot of Numeric Features (Outlier Check)



Observation: total_rooms, total_bedrooms, and population show several points above the upper whisker, reflecting genuinely high-density districts rather than data errors.

Chart 2: Histogram of Median Income (Univariate)



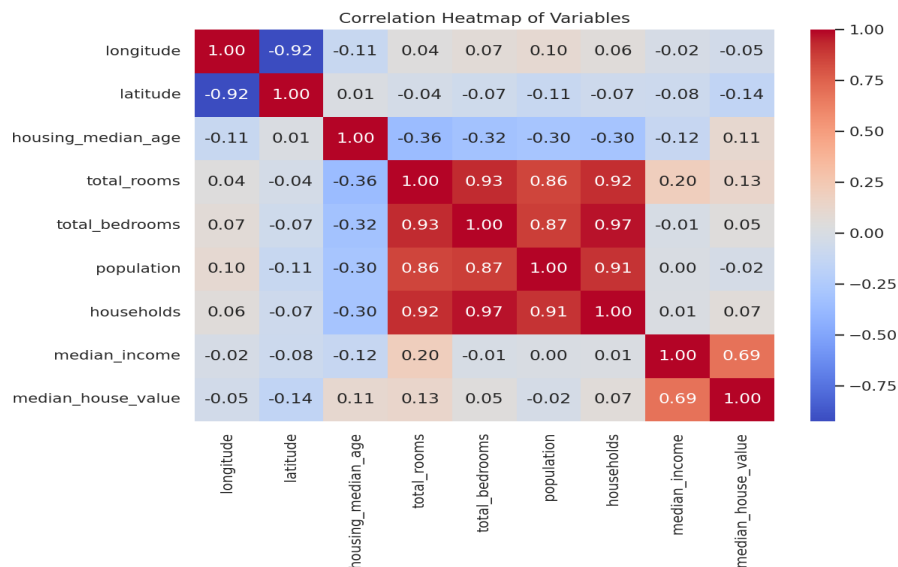
Observation: Median income is right-skewed, with most districts clustered between roughly 2 and 5 (tens of thousands of dollars).

Chart 3: Median Income vs Median House Value (Bivariate)



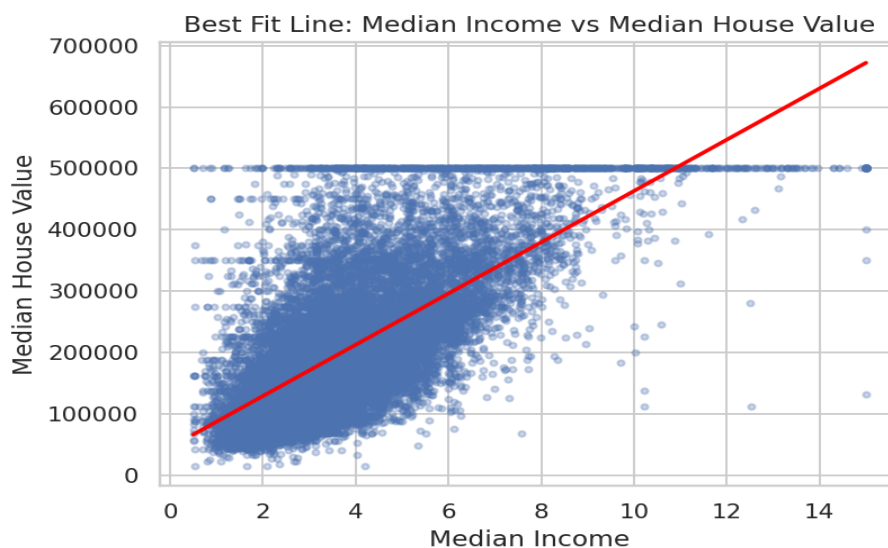
Observation: A clear positive relationship is visible - districts with higher median income tend to have higher median house values.

Chart 4: Correlation Heatmap



Observation: median_income has the strongest positive correlation with median_house_value. total_rooms, total_bedrooms, population, and households are all highly correlated with each other.

Chart 5: Best Fit Line - Median Income vs Median House Value



Observation: The fitted regression line confirms a positive linear trend between income and house value, though the spread of points shows income alone does not fully explain price variation.

Model Building and Evaluation

After encoding `ocean_proximity` and separating features (X) from the target (y), the data was split 80/20 into training and test sets (`random_state=42`) and scaled using `StandardScaler`. A Linear Regression model was trained on the scaled training data and evaluated on the held-out test set.

Metric	Value
Mean Squared Error (MSE)	5,066,374,909.74
Mean Absolute Error (MAE)	51,829.49
R-squared (R2)	0.6134

The model explains about 61.3% of the variance in median house value ($R^2 = 0.613$). On average, predictions are off by roughly \$51,829 (MAE).

Prediction on unseen data:

For a sample unseen district (longitude -122.25, latitude 37.85, `housing_median_age` 30, `total_rooms` 2000, `total_bedrooms` 400, `population` 1000, `households` 350, `median_income` 5, `ocean_proximity` NEAR BAY), the model predicted a median house value of approximately \$269,812.

Model coefficients and intercept:

Intercept: 207,194.69. The largest positive coefficients belong to `households` and `median_income`, while longitude and latitude carry the largest negative coefficients - consistent with price falling as districts move further from the coastal reference point used in scaling.

7. Key Findings

- Median income is the strongest driver of median house value, both by correlation (Chart 4) and by regression coefficient magnitude.
- `total_rooms`, `total_bedrooms`, `population`, and `households` are highly correlated with each other, meaning they carry overlapping rather than independent information (Chart 4).
- Most districts fall in the '<1H OCEAN' and 'INLAND' categories (76% combined); 'ISLAND' has only 5 records, too few to reliably estimate its effect on price.
- The linear regression model achieves an R-squared of 0.613, explaining roughly 61% of the variance in house prices - a reasonable baseline for a linear model.
- The boxplot (Chart 1) shows genuine right-skew, not data errors, in `total_rooms`, `total_bedrooms`, and `population`, which a linear model does not correct for.

Limitations:

- The ISLAND category has only 5 records, too few to draw reliable conclusions about it.
- The dataset reflects 1990 census data, so absolute price levels do not reflect current housing markets, though relative relationships between features likely still hold directionally.
- A purely linear model cannot capture non-linear effects (e.g. diminishing returns on income), which limits accuracy at the extremes of the price range.

8. Conclusion

This project applied a full linear regression workflow - data cleaning, exploratory analysis, feature encoding, model training, and evaluation - to the California Housing dataset. The model achieved an R-squared of about 0.61, confirming that income and location are strong predictors of district-level house prices, while some variation remains unexplained by a purely linear approach. The main challenge was deciding how to handle the missing values in `total_bedrooms`; median imputation was used to preserve the full dataset. Future work could explore non-linear models (e.g. Random Forest, Gradient Boosting) or engineered ratio features (e.g. rooms per household) to improve accuracy.